

Tribhuvan University
Faculty of Humanities and Social Sciences
OFFICE OF THE DEAN
2024
Bachelor of Computer Applications

Course Title: DotNet Technology
Code No: CACS302
Semester: Fifth Semester

Group B

Attempt any SIX questions.

2. What is managed code? Give reason to justify the statement "DOT NET framework is platform independent an Language interoperability".
3. What is optional parameter? Write a C# program which stores values in two enumerations, Department and College. It uses two functions to display the data contained in Department and College enumerations.
4. What is difference between value type and reference type in C#? Explain the concept of boxing and unboxing with suitable program.
5. What is interface? Explain with suitable program, how interface can be used to perform multiple inheritance?
6. What are use of finalizer in C#? Explain with suitable program, how base keyword is used to call member function and constructor of parent class?
7. What are keyword used in exception handling? Write a C# program to handle IndexOutOfRangeException and InvalidCastException.
8. What is query syntax in LINQ? Explain the standard query operators "select", "contains", "orderBy" and "where" in LINQ with suitable example.

Group C

Attempt any TWO questions.

- 9.a) What is an event? Explain with suitable program, how event is handled in C#.
- b) What is Lambda expression? Explain the concept of generic delegates with suitable program.
10. How this keyword is used to declare indexer? Write C# program to overload unary (++) and relational(==) operator.
11. What are component of [ASP.NET?](#) Explain various validation server control in [ASP.NET](#) with suitable program.

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Practical Examination 2024

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Semester: V

Full Marks: 20
Pass Marks: 8
Time: 1.5 hrs.

Set-A

Attempts any THREE questions

5. Write a C# console application to define **Person** class and derive **Student** and **Employee** Class from it to demonstrate hierarchical inheritance.
6. Write a C# program to define class **Distance** with attribute **feet and inches** and then add and subtract the two distances using binary operator overloading.
7. Write a C# program to display Employee records in descending order with respect to employee name using LINQ.
8. Write a c# program to connect to the database and insert 5 student records and display the student record from College database whose faculty is "Computer Application". (Database: College Table: Student (Roll, Name, And Faculty))

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Set-B

Attempts any THREE questions

1. Write a C# program to define class **Employee** with attribute **Eno, Ename, Address, and Salary**. Set default and user-defined value for these attribute using Constructor and then display values of these attribute.
2. Write a C# program that will read balance and withdraw amount from keyboard and display the remaining balance on screen if balance is greater than withdraw amount otherwise throw an exception with appropriate message.
3. Write a C# program to select Students who are lived in **Kirtipur** and studied in **RR Campus** using LINQ
4. Write a c# program to connect to the database and insert 3 Employee records and delete the Employee record from company database who's Department is "Sales".
(Database: Company Table: Employee (EId, Ename, Salary, Department))